

The neurodevelopmental role of serotonin in the neonatal rat.

Does it matter that your mother has a genetic reduction in serotonin transporters

by

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Abstract

Many neurobiological processes are mediated by serotonergic signalling. Serotonin has been shown to increase neurogenesis in the adult brain. Here we investigated the developmental role of serotonin in the neonatal rat using serotonin transporter knockout rats. Measuring body weights at post natal day 7 revealed homozygous rats to be lighter than wildtype and heterozygous rats. Furthermore offspring from homozygous mothers were lighter than offspring from heterozygous mothers, indicating that maternal serotonin also plays a role in body weight. Neurogenesis was investigated through immunohistochemistry which revealed a positive relationship between the amount of expected extracellular serotonin in the brain between the different genotypes and neurogenesis. The relationship between serotonin and brain development might explain the heightened predisposition to psychiatric disorders in subjects with excessive extracellular serotonin during their lifetime. Next to analysing the neurodevelopmental role of serotonin some recommendations for the Golgi-Cox method are provided for future study.

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Introduction

Serotonin (5-HT) is a widely studied neurotransmitter that mediates many diverse behaviours such as mood, sleep, memory, temperature, appetite, sexual behaviour, movements, gastrointestinal (GI) motility, and many more (Booij, Richard, Szyf, & Benkelfat, 2015; Calabrese et al., 2013; Kepser & Homberg, 2015; Stahl, 1998). In this study the focus lies on the relationship between serotonin and neurodevelopment and the effect this could have on neuropsychiatric disorders such as anxiety and depression.

In the brain extracellular 5-HT levels are increased upon activation of serotonergic neurons from the raphe nuclei (RN) that project to almost all parts of the brain. 5-HT homeostasis is primarily regulated by the Na⁺/Cl⁻ dependent serotonin transporter (SERT). SERT reuptakes 5-HT from the extracellular space into the presynaptic serotonergic nerve terminal where 5-HT can be recycled for future release (Homberg et al., 2007; Murphy & Lesch, 2008).

Treatment of neuropsychiatric disorders usually involves selective serotonin reuptake inhibitors (SSRIs) to reduce symptoms. SSRI's inhibit the 5-HT reuptake by blocking the serotonin transporter leading to excessive extracellular 5-HT in the synaptic cleft. 5-HT subsequently interacts with various serotonin receptors in the synapse which can reduce anxiety and depression (Stahl, 1998).

Surprisingly subjects with a genetic reduction in serotonin transporters (also leading to enhanced extracellular 5-HT) are known to suffer from heightened anxiety-related personality traits (neuroticism), stress-sensitivity, and increased susceptibility to mood disorders and depression (Altieri et al., 2014). One of the explanations for this counterintuitive observation might be the effect excessive levels of 5-HT have during (neonatal) development.

5-HT, besides influencing brain functions as a neurotransmitter, has been shown to have a developmental role in the brain as a neurotrophic factor (Wei Yan, Wilson, & Haring, 1997). Different 5-HT receptors, acting at different developmental stages modulate developmental processes such as neurite outgrow, neurogenesis, gliogenesis, apoptosis, axon branching and dendritogenesis (Gaspar, Cases, & Maroteaux, 2003; Kepser & Homberg, 2015; Trowbridge, Narboux-Nême, & Gaspar, 2011). These processes start taking place before birth (Booij et al., 2015) and SERT is expressed much more broadly during development than in adulthood (Calabrese et al., 2013; Gaspar et al., 2003). Serotonergic neurons are amongst the earliest neurons to develop and 5-HT is released by growing axons even before conventional synapses are established (Gaspar et al., 2003). However before the brain synthesises 5-HT itself, 5-HT is derived from the mother through the placenta (Bonnin & Levitt, 2011). The differences in neurological development due to different levels of 5-HT may cause behavioural differences and neuropsychiatric disorders later in life.

Since 5-HT is a monoamine that is subject to maternal transmission by crossing the placenta (Altieri et al., 2014; Kepser & Homberg, 2015; Rampono, Proud, Hackett, Kristensen, & Ilett, 2004; Trowbridge et al., 2011), mothers with an increased extracellular 5-HT level can pass this high level of 5-HT to their offspring (Witteveen et al., 2013). This is of importance since it can have multiple effects, for instance it has been shown that serotonergic antidepressants during pregnancy and breastfeeding can change brain development in offspring (Kepser & Homberg, 2015). This effect can result from these antidepressants passing through the blood placenta barrier and thus directly affecting 5-HT in the foetus, or from high levels of 5-HT from the mother that cross the placental barrier.

In this study changes in the neonatal brain due to differences in levels of 5-HT were investigated in rats which have a genetic alteration in serotonin transporter knock-outs. Homozygote serotonin transport knock-out rats have a 100% reduction in serotonin transporters, leading to excessive extracellular 5-HT compared to normal wild type rats. Heterozygote rats only have a 50% reduction in serotonin transporters, and subsequently much lower extracellular 5-HT compared to homozygous rats (Homberg et al., 2007). Many studies have shown that, in adulthood 5-HT increases neurogenesis especially in the hippocampal formation (Alenina & Klempin, 2015; Benninghoff et al., 2012). Therefore hippocampal neurogenesis was studied here to establish the neurotrophic role of 5-

HT in pre and neonatal development. Revealing differences between the genotypes could provide useful information since the hippocampus is a brain area where 5-HT plays an important role and is a key region in depression and antidepressant therapy (Fanselow & Dong, 2011). In addition, 5-HT can affect the morphological structure of neuronal cells, and especially the outgrowth of dendritic spines in adulthood (McEwen, 2007).

Offspring with a wild type genotype were hypothesized to have the least neurogenesis, heterozygous rats were hypothesized to have slightly higher levels of neurogenesis and homozygous rats were hypothesized to have the most neurogenesis. Furthermore offspring of HET mothers which have a 50% reduction in serotonin transporters were hypothesized to have lower levels of neurogenesis than offspring from rats with a 100% reduction in serotonin transporters since mothers can transfer 5-HT to their offspring. Next to neurogenesis the effect of 5-HT on dendritic spine density was set out to be investigated in this study, however this was not possible due to time limitations and procedural setbacks. A few recommendations for a future study in which dendritic spine density will be investigated are given.

Since a substantial group of humans have a short variant of the polymorphism (HTTLPR) leading to reduced transcriptional efficiency of the SERT gene (Lesch et al., 1996) which predisposes them to psychiatric disorders, and the 5-HT system is comparable between rats and humans, these results can contribute to better understanding the role of 5-HT in brain development and its link to psychiatric disorders.

Materials and methods

Animals:

Wistar strain rats were bred which either had a wild type (SERT^{+/+}), a heterozygote (SERT^{+/-}) or a homozygote (SERT^{-/-}) knock-out genotype of the serotonin transporter. Wild type (WT) animals have no reduction in the serotonin transporters, heterozygote animals have a $\pm$ 50% reduction in the serotonin transporter (further referred to as HET) and homozygote animals have a 100% reduction in SERT (further referred to as HOM). Animals were housed in polycarbonate cages in a temperature controlled room at 21°C and 55% humidity with a controlled 12 h light/dark cycle with onset of light at 7am. Animals had access to food and water ad libitum.

In total the offspring of 9 litters of 5 combinations of genotypic parents were used (table 1, table 2) for neurogenesis and spine density. To this extend brains of 4 pups per litter were used for immunohistochemistry and brains of 4 pups per litter were used for histology. Only heterozygote SERT KO mothers and homozygote SERT KO mothers were used for breeding since HET mothers have almost similar extracellular 5-HT levels as WT rats while HOM SERT KO mothers have increased 5-HT levels (Homberg et al., 2007) which could further affect the pups' brain development.

Number of litters	Genotype Male	Genotype Female	Probability Genotype offspring
2	WT	HET	50% WT / 50% HET
2	HET	HET	25% WT / 50% HET / 25% HOM
3	HOM	HET	50% HOM / 50% HET
1	HET	HOM	50% HET / 50% HOM
1	HOM	HOM	100% HOM

Table 1. Overview of number of litters used per pairing combination with probabilities of resulting genotypes of offspring.

Genotype mother	Genotype pup	Animals used for neurogenesis	Animals used for spine density
HET	WT	7 (3♀, 4♂)	4 (1♀, 3♂)
	HET	15 (10♀, 5♂)	18 (11♀, 7♂)
	HOM	7 (2♀, 5♂)	6 (1♀, 5♂)
HOM	HET	2 (2♂)	3 (2♀, 1♂)
	HOM	6 (3♀, 3♂)	5 (3♀, 2♂)
Total		37	36

Table 2. Overview of number of animals used in the experiments for neurogenesis and spine density.

Cell proliferation/ neurogenesis

Cell proliferation was assessed through bromodeoxyuridine (BrdU) incorporation into nuclei during the S phase of the cell cycle (Taupin, 2007; Zhao et al., 2014). BrdU is a nucleoside that substitutes thymidine during DNA replication in proliferating cells. Antibodies against BrdU were used to detect cells that were actively replicating their DNA. Pups were injected intraperitoneal with BrdU (25mg/kg) at post-natal day 7, decapitation and brain removal occurred two hours after injection. A tail clip was taken after decapitation to assess genotype. Brains were postfixed overnight in a 4% paraformaldehyde solution at 4°C before placed in a 30% sucrose in 0.01 M PB solution to cryoprotect. Brains were then quick-frozen in isopentane at – 40°C and stored in a freezer at -80°C. Frozen coronal sections (30 µm) were made on a freezing microtome (set at -21°C) and collected in six parallel series in 0.1 M PBS containing 0.1% azide. One series per rat was used for staining. Brain slices were mounted on gelatine coated glass microscope slides. Immunoreactivity for incorporated BrdU was retrieved by pouring boiling 10 mM sodium citrate buffer (pH 6.0) on brain sections refreshing the citrate buffer every 3 minutes for a period of 10 minutes, followed by placing the brain sections in 2 N HCl at room temperature for 90 minutes. Brain slices were then washed in PBS, blocked in 1% BSA (Sigma) for 1 hour at room temperature and washed with PBS again. Incorporated BrdU was detected by overnight incubation with an anti-BrdU mouse monoclonal antibody conjugated to biotin (1:2000, ab2284, Abcam). Sections were washed in PBS, and then incubated for 1 h at room temperature with avidin biotin peroxidase complex (1:1000, Vectastain). The location of BrdU epitopes was revealed by DAB staining and tissue was counterstained with neutral red before coverslipped. No DAB staining was seen in sections from animals not injected with BrdU or for brain slices that were not treated with BrdU antibody, confirming the specificity of the reaction (adapted from (Steel et al., 2014)). Stained cells were counted in 3 region of the hippocampus proper: the dentrite gyrus (DG), coronus ammonis 1 and 2 were combined (CA1/2) and coronus ammonis 3 (CA3).

Dendritic spine density

Histology was performed using the Golgi-Cox method. Pups were decapitated at post natal day 7. A tail clip was taken after decapitation to assess genotype. Brains were quickly removed and quick frozen in isopentane at – 40°C during 3 minutes. Brains were stored in a freezer set at -80°C until treated with the Golgi-Cox solution (appendix). Brains were placed in a conical tube that blocked light from getting though containing at least 5 ml of Golgi-Cox solution per cm² of tissue. After 24 hours the Golgi-Cox solution was refreshed, brains were left in the fresh solution for 14 days after which being transferred in a 30% sucrose solution for at least 3 days. Coronal sections (100 µm) were made using a cryostat where brain slices were immediately mounted on a glass microscope slide. The revelation was done by emerging brain sections in multiple chemical solutions. First sections were washed in distilled water for 1 minute, sections were then moved to a 17.5% ammonium hydroxide solution for 30 minutes, washed in distilled water for 1 minute and placed in a 1% sodium thiosulfate solution for thirty minutes. After washing in distilled water for 1 minute sections were placed in successive bathes containing increasing percentages of ethanol (50%, 70%, 95%) for 1 minute then two bathes of 100% ethanol for 5 minutes each. After being submerged in two bathes of histoclear during 15 minutes each, sections were coverslipped. Due to difficulties executing the protocol multiple test trails were performed. Different parameters were tested for the best results. Different thickness of brain sections (80 µm, 100 µm and 150 µm) were sliced on a microtome vs. a cryostat, mounted on superfrosted microscope slides vs. gelatine coated microscope slides, mounted using a 6% sucrose solution vs. not using a sucrose solution. Tissue was tested that had been in the Golgi-Cox solution for 7 day, 10 days or 14 days which had been postfixed overnight in 4% formaldehyde before

placing in the Golgi-Cox solution vs. fresh tissue. The method which gave the best result is discussed in the results section with additional information in the implications.

Statistical analysis

Statistical analysis was performed using SPSS (version 22). Significance was assumed when the p value was <0.05 . Data are presented as means $\pm$ standard error of mean. Data was analysed with two-way ANOVA comparing body weights and number of cells. Post-hoc tests were used to identify the source of statistical differences where appropriate.

Results

Body weights:

Univariate two way analysis of variance showed main effects of genotype mother ($F_{1,44}=5.110$; $p=0.029$) and genotype pup ($F_{2,44}=6.599$; $p=0.003$) on bodyweight, no interactions were shown.

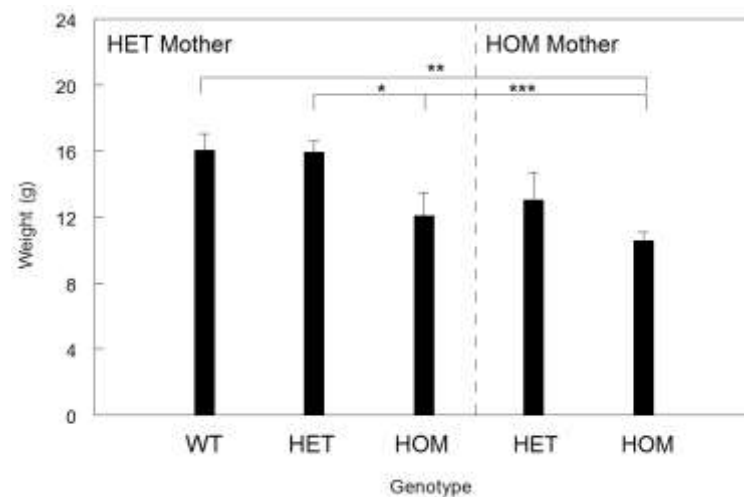


Figure 1. Pups' mean $\pm$ S.E.M. body weight (g) at postnatal day 7. Pups' weights are shown per genotype having either a HET mother (shown on the left side of figure) or HOM mother (shown on the right side of figure).

Body weights differ significantly between groups ($F_{4,44}=8.102$, $p<0.001$). HOM pups were lighter than rats with a HET ($p<0.001$) and WT ($p=0.001$) genotype while WT did not differ from HET significantly ($p=.791$). Furthermore pups from HOM mothers were lighter than pups from HET mothers ($F_{1,44}=15.142$, $p<0.001$.)

Cell proliferation/ neurogenesis

To assess if maternal levels of 5-HT were of influence in neurogenesis next to genotype, offspring of HET mothers were compared to offspring of HOM mothers (figure 2). No significant differences were found for the total number of BrdU labelled cells in the observed areas of the hippocampus between groups ($F_{4,34}=1.388$, $p=.262$).

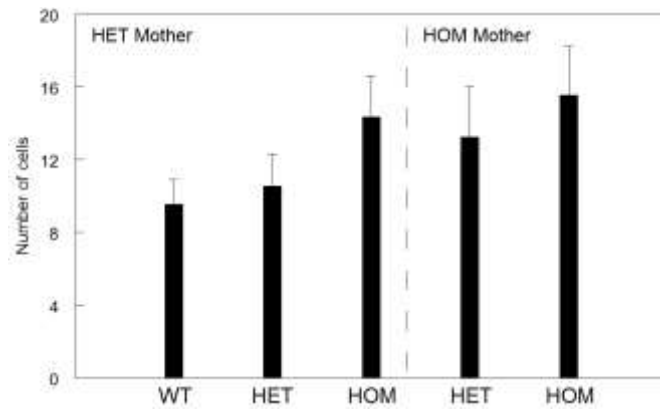


Figure 2. Mean $\pm$ S.E.M. BrdU labelled cells in the areas DG, CA1/2, CA3 combined, split for genotype pups from either a HET mother (shown on the left side of figure) or HOM mother (shown on the right side of the figure).

Given the relatively small number of animals per group, the data were combined for both maternal genotypes. Looking at the total number of BrdU labelled cells in the observed areas of the hippocampus, no significant differences between the genotypes were found, however a trend can be seen ($F_{2,34}=2.640$, $p=0.087$) (figure 3).

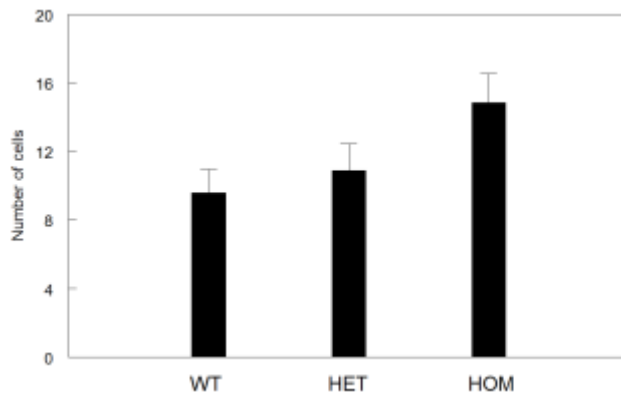


Figure 3. Mean $\pm$ S.E.M. BrdU labelled cells in the areas DG, CA1/2, CA3 combined, split for genotype pups.

Splitting the hippocampus in 3 regions led to a significant difference in number of cells in the CA3 ($F_{2,33}=3.904$, $p=.031$) (post-hoc, HOM differ from WT animals ($p=.034$)) of the hippocampus but not in the CA1/2 ($F_{2,34}=2.105$, $p=.138$) and DG ($F_{2,34}=1.540$, $p=.230$) (figure 4).

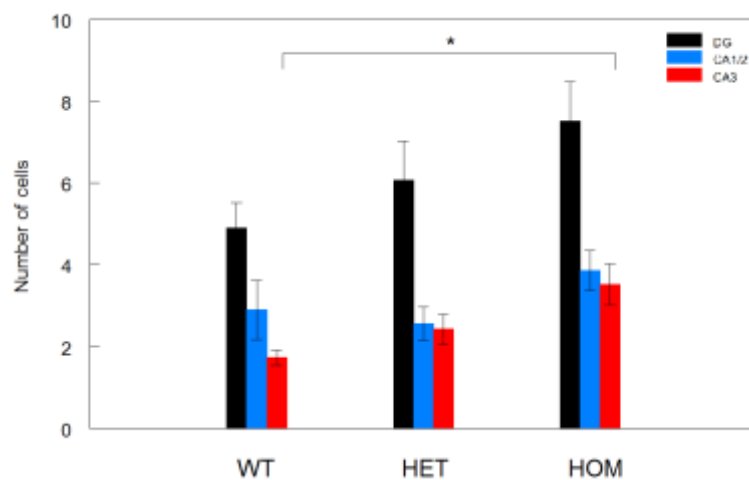


Figure 4. Mean $\pm$ S.E.M. BrdU labelled cells in different regions of the hippocampus (dentate gyrus, CA1/2 and CA3), split for genotype pups.

No correlation was found between body weight and hippocampal size (data not shown).

Dendritic spine density

No data about spine density can be presented here because of the limited time remaining after the multiple test trails. Instead data regarding the best parameters to be used in future brain histology is shown.

Tissue stayed mostly intact when cut on a cryostat. Mounting the tissue after being cut on the microtome was nearly impossible since tissue defrosted and became pulpy or crumbled upon touching by a brush. On the cryostat moving tissue over the blade too fast resulted in multiple cracks in the tissue, cutting on very low velocity gave the best results. Postfixing the brains in 4% formaldehyde before emerging in the Golgi-Cox solution did not prevent cracking or crumbling when slicing the brains. The number of days the brains were in the Golgi-Cox solution had no noticeable effect on cracking and crumbling but did affect the visibility of the dendrites (figure 5). Sections cut at 150 μm started to crumble the most and folds in the tissue appeared when slicing. No noticeable difference between 80 μm and 100 μm thick sections were observed on cracking and crumbling. Both tissue mounted on superfrosted glass slides and gelatine coated glass slides came off during emerging in the several solutions during the revelation process. After revelation the number of visible dendrites was still small and mostly only partly visible around the cell body, dendritic spines were hard to distinguish at 400 times magnification.

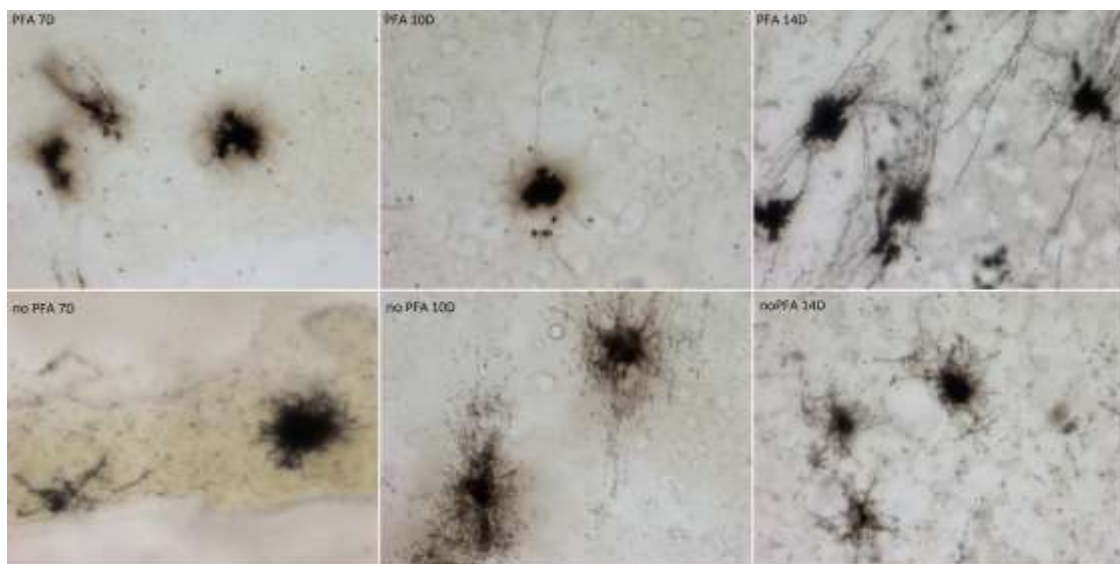


Figure 5. Stained cell bodies and dendrites for brain tissue after 7, 10 and 14 days of Golgi-Cox solution. Brain tissue postfixing in formaldehyde (top row) vs. brain tissue that was not postfixing in formaldehyde (bottom row) (400 times magnification).

Dendritic spines were best observable after tissue had been in the Golgi-Cox solution for 14 days, postfixing tissue did not counteract the staining (figure 5).

Discussion

In the present study differences in developmental processes caused by differences in genotypes regulating the serotonin transporter were analysed. Differences were found for body weight and for the different genotypes in neurogenesis for different regions of the hippocampus, especially in the CA3 region.

5-HT appeared to have a complex effect on body weight at post natal day 7. First of all, HOM pups were significantly lighter than HET and WT pups. This indicates there might be a relationship between extracellular 5-HT and body weight, where more extracellular 5-HT results in lighter body weight. Furthermore HOM mothers had significantly lighter pups than HET mothers. There are at

least two alternative explanations for these findings. Firstly the high levels of 5-HT in HOM mothers may have transferred to their pups during pregnancy, since pups are dependent on maternal 5-HT before synthesising 5-HT themselves. This high level of 5-HT may have had an effect on body weight through regulation of the endocrine system (Yadav et al., 2009). 5-HT could for instance be influencing the sleep pattern and appetite of the pups leading to differences in body weight. Alternatively the reduced body weight could be due to reduced maternal care after birth by HOM mothers. Similar to our findings a study by Angoa-Pérez et al. showed that mothers with a complete depletion of 5-HT level show less maternal care and pups with a complete depletion of 5-HT received less maternal care than their siblings, this indicates that genetic alterations in the 5-HT system plays an important role in maternal care (Angoa-Pérez et al., 2014). Testing this hypothesis would require cross fostering to compare different genotypes of offspring of a HET mother receiving maternal care from HOM mothers to offspring of a HOM mother receiving maternal care from HET mothers. As both hypothesis are not mutually exclusive, a combination of these two effects is also possible. Establishing these factors related to body weight is a topic for future research.

The results show that extracellular 5-HT has a positive effect on neurogenesis, although this did not always reach statistical significance, WT pups show the least neurogenesis in all areas of the hippocampus while HOM pups show the most neurogenesis. This correlates with the extracellular 5-HT levels expected to be present in the brain of these pups, where WT have the lowest level of extracellular 5-HT and HOM have the highest level. HOM pups show significantly more neurogenesis in the CA3 area of the hippocampus than WT. While the different regions of the hippocampus have distinctive tasks this is probably not the reason why significance is only reached in this area. A more likely explanation is that neurogenesis is elevated in all areas of the hippocampus in HOM pups but only shows significance for the CA3 areas since this area had the lowest variation between groups.

This positive effect of extracellular 5-HT on neurogenesis shows resemblance with the effects SSRIs have on neurogenesis (Alenina & Klempin, 2015), this indicates that serotonin indeed acts as a neurotrophic factor. It has been shown that SSRIs in adults lead to an increase in brain-derived neurotrophic factor (BDNF) (Homberg, Molteni, Calabrese, & Riva, 2014) which is thought to be a mechanism whereby 5-HT could induce neurogenesis. However HOM rats show a decrease in BDNF at PN day 7 compared to wildtype pups (Calabrese et al., 2013). Since HOM rats show an increase in neurogenesis despite having less BDNF, BDNF expression might not be as essential for neurogenesis as 5-HT.

The effect of higher levels of 5-HT during critical windows of brain maturation in pre and postnatal development may eventually lead to the heightened predisposition to psychiatric disorders (Calabrese et al., 2013), especially since pups rely solely on their mothers 5-HT in the first half of pregnancy in which brain development is high.

There are also some preliminary data to suggest that the genotype of the mother has an influence on the neurogenesis in the pups. Both HET and HOM pups from HOM mothers showed more BrdU positive cells than animals of the same genotype from HET mothers. However this difference was not significant and more animals are needed to have enough statistical power for comparison. In this study we originally set out to compare WT male/HET female, HOM male/HET female and HET male/HOM female combinations. However due to several mix ups in pairing breeding animals, some HET male /HET female and HOM male/HOM female pairs were inadvertently included. Since the genotypes of the pups were not known until days after brain removal some groups were relatively small. For instance only 2 HET animals from a HOM mother could be used to investigate neurogenesis in this experiment making it impossible to do comparative analysis with other groups. Having more animals in this and other groups might have led to more significant results, supporting the hypothesis that transferred 5-HT from the mother can have great effects on their offspring. Looking at the different brain regions, the dentate gyrus showed more newly formed cells than the other regions of the hippocampus. This was expected since newly formed cells migrate outwards from the sub ventricular zone near the dentate gyrus (Altman & Bayer, 1990; Kempermann, Song, & Gage, 2016). There was no genotype differences with respect to the percentage of cells formed per hippocampal region between groups (data not shown).

This study has shown a pathway in which SERT reduction might increase the risk for depression, anxiety disorders and autism through a genetic role, however since differences in brain development can only partly explain this relationship and genetic effects can be amplified by gene-environment and gene-gene interactions (Murphy & Lesch, 2008), further research should focus on the role non-genetic environmental factors and maternal care have, including behavioural test in offspring to see if the genotype of the mother is of importance for behavioural changes in the offspring next to their own genotype. This would lead to further understanding of how environmental and genetic factors influence brain and behaviour.

Implications

Future breeding of SERT KO rats has to be carefully monitored because of genotypic maternal effects that may occur. Offspring from HOM rats can be different from offspring from HET rats in 5-HT levels and behaviour. These differences could be confounding factors which must be included in all experiments done with these genetic altered animals. To this extent genotypes of the mother of the animals that are used in the experiments should be included.

In a future study the dendritic spine density can be looked at after brains have been treated with the Golgi-Cox method according to the tested protocol in this study. Including spine density next to neurogenesis in the SERT KO rats would provide more information about the effect of serotonin and maternal serotonin on brain development. Previously Yan, Wilson & Haring have shown that 5-HT depletion in neonatal rats leads to a decrease in dendritic spine density and granule cells (W Yan, Wilson, & Haring, 1997) raising the question if excessive extracellular 5-HT will increase dendritic spine density. When this study is performed brains should be in the Golgi-Cox solution for two weeks, sliced very slowly on a cryostat set at around -14 for both the chamber temperature as the cryostat itself. If crumbling occurs temperature can be increased slightly, if the tissue sticks too much temperature can be decreased slightly. The microscope glass can be placed over the tissue to mount the tissue on the glass gently making sure the tissue is not pushed against the freezing plate of the cryostat, this will make the tissue stick to the glass. After the tissue is on the glass, heat can be applied to the tissue using your thumb on the other side of the glass, this will make the tissue stick better. If some folds appear tissue can be pushed down with a brush very gently when mounting. The tissue should be stuck to the glass slide as tightly as possible to prevent falling off during revelation. During revelation glass slides containing sections should be transferred to the different solutions with great care and consideration to prevent tissue from separating from the glass slide. When looking at the dendritic spines through the microscope magnification should be increased to 1000 times to make distinguishing the different spines easier.

Levels of 5-HT in postnatal as well as prenatal rats should be assessed to establish whether HOM rats indeed have elevated levels of extracellular 5-HT and especially to see how much maternal 5-HT is transferred through the placenta by HET compared to HOM mothers.

Although no correlation was found between body weight and hippocampal size this should be more closely investigated in future research by including brain weight next to size.

Conclusion

Overall this study indicates that the difference in 5-HT homeostasis is likely to have an effect on neurogenesis and body weight of the pup. Extracellular 5-HT of the mother seems to be an important factor too since 5-HT is transferred from mother to offspring. Further research has to be done to determine possible differences in maternal care between the different SERT KO genotypes and the effects these might have on behaviour and physiopathology of psychiatric diseases in later life.

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Appendix:

Golgi-Cox solution (100ml) (Protected from light at all times):

Mix 5% Mercury (II) chloride solution (20ml) + (HgCl₂) 5% potassium dichromate solution (20ml) + (K₂Cr₂O₇) 1.67% solution of potassium chromate (K₂CrO₄) (60ml) in a conical tube. Leave the solution in a dark place at room temperature for 24 hours. After 24 hours decant the solution into another tube avoiding decanting of precipitate.